

Make Your Own USB Data Acquisition System

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Presented here is a data acquisition (DAQ) system that can be used to interface six sensors and four digital-to-analogue channels. The acquired voltage levels can be displayed on a personal computer (PC). The DAQ device may be calibrated

for precision and benchmarked with various standards, and delivered as an inexpensive USB DAQ product for hobbyists and students.

Irrespective of our stream of engineering or science, most of us do programming and data analysis on computers due to ease in data handling, storage and manipulation.

This project may be helpful for those who want to develop an integrated system that consists of a DAQ and other system circuitry in a single unit. It would provide a greater insight to the design and development of a USB DAQ system itself! The article includes details of circuit designs, software development for user-interface using C# and LabView, printed circuit board (PCB) development and test results.

It is assumed that the

reader is familiar with basic 'C' programming and AVR programming styles. However, best efforts have been made to make the contents logical enough to help even a beginner understand the project and tweak with the codes.

DAQ systems

A DAQ system acquires data from a sensor mounted on a physical system and brings it to your PC.

Typically, the output from a sensor is available in the form of a voltage signal. For example, the output from a commonly used temperature sensor like LM35 is 10mV/°C under normal environmental conditions, which var-

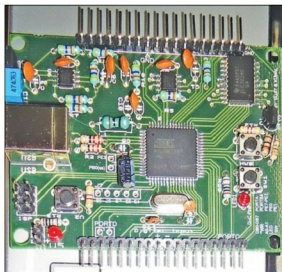


Fig. 1: Author's prototype

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	2.5V
TP3	Low when S2 pressed

TABLE I
Specifications of the Developed DAQ Device

Feature	Specifications	Remarks
Digital inputs/outputs (I/Os)	16 reconfigurable I/Os, Maximum: 40 mA	Each of the 16 pins can be configured as input or output
Analogue inputs	6 (8-bit resolution), Maximum: 5V	Can be extended up to 8
Analogue outputs	4 (8-bit resolution), Maximum: 5V, 0.45 mA	Can be increased by adjusting the op-amp gains and using an external supply
Counters	Any digital I/O can be used as counter (Maximum: 4 MHz using external interrupt on Port-D 0-3)	To be implemented in PC software or by modifying the firmware
Interface	GUI may be developed using any programming language with DLL interfacing	Developed and tested with NI LabView and Microsoft Visual C#
Driver for programming	Libusb 1.0 (Free) (http://www.libusb.org/)	Installs automatically on installing ATMEL Flip programmer (free)
Application programming interface (API) driver	ATUSBHID.DLL	Free-required for communication
Upgrade	Open source firmware written using LUFA	Board can be reused for multiple purposes
Sampling frequency	9.6 kilo-samples-per-second with Generic USB HID implementation	One can use a different USB class with full-speed USB data transfer rate of 12 Mbps

Simply plug the device on the USB port and the software gets installed in Generic HID Class